Game-Theory-Based Balance: Formalizing the Machine Unlearning Problem as a Game Between Model Provider and Data Provider

PhD Proposal Submitted to

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

By

Abhishek Hupele
Research Scholar-240900465
School of Computer Engineering,
Manipal Institute of Technology, Manipal

Under the Supervision of

Dr. Sanjay Singh
Professor (Guide)
School of Computer Engineering,
Manipal Institute of Technology, Manipal

Dr. P.C. Siddalingaswamy
Associate Dean (Co-Guide)
School of Computer Engineering,
Manipal Institute of Technology, Manipal

Dr. Naveen Kulkarni
Director – Product Engineering - AI (Co-Guide)
Edgeverve Systems Limited

March 2026

Game-Theory-Based Balance: Formalizing the Machine Unlearning Problem as a Game Between Model Provider and Data Provider

Abhishek Hupele

Research Scholar-240900465

School of Computer Engineering, Manipal Institute of Technology, Manipal

March 2026

Supervision

- Dr. Sanjay Singh, Professor (Guide), School of Computer Engineering, MIT Manipal
- Dr. P.C. Siddalingaswamy, Associate Dean (Co-Guide), School of Computer Engineering, MIT Manipal
- Dr. Naveen Kulkarni, Director – Product Engineering - AI, EdgeVerve Systems Limited

Contents

List of Figures

List of Tables

Glossary

AI	Artificial Intelligence
CNN	Convolution Neural Network
DNN	Deep Neural Network
LLM	Large Language Model
LSTM	Long Short Term Memory
MIA	Membership Inference Attack
ML	Machine Learning
MUL	Machine Unlearning
NPTEL	National Programme on Technology Enhanced Learning
PCA	Principal Component Analysis
PII	Personally Identifiable Information
RAI	Responsible AI
RNN	Recurrent Neural Network
SISA	Sharded, Isolated, Sliced, Aggregated
SVM	Support Vector Machine

Chapter 1

Introduction

1.1 Introduction to Machine Unlearning

Machine Unlearning means making a machine forget specific training data so that it behaves as if the data was never used. Reasons include withdrawal of consent, unethical acquisition, adversarial data, copyright issues, or compliance requirements. Challenges: stochasticity, incrementality, catastrophe of unlearning. Main approaches:

- Centralized Unlearning
- Federated (Distributed) Unlearning
- Unlearning Verification
- Privacy and Security Issues

1.2 Introduction to Game Theory

Game theory studies strategic interactions between players. Key concepts: players, strategies, payoffs, information, equilibrium (Nash Equilibrium). Applications span economics, political science, biology, computer science, business, and social sciences.

1.3 Intersection of Game Theory and Privacy

Game theory helps balance privacy and utility. In machine unlearning, over-unlearning harms performance, under-unlearning risks privacy leaks. This proposal formalizes unlearning as a game between model provider and data provider.

1.4 Problems Identified

- PII misuse (Cambridge Analytica, Apple Card bias, Amazon hiring bias, Clearview AI, Target pregnancy prediction, Gartner's GenAI misuse forecast).
- Need for balance between privacy and utility.
- Technical difficulty of certified removal.
- Lack of universal unlearning frameworks.
- Performance degradation and leakage risks.
- Variability in models, modalities, adversaries.

1.5 Proposed Solutions

Formalize machine unlearning as a game-theoretic problem. Develop generalized frameworks applicable across ML models, modalities, and adversary settings. Flexible scope: start with classical ML and neural networks, expand to transformers and federated settings.

Chapter 2

Literature Review

Sl No	Title	Authors	Year	Methodology	Result
1	Responsible AI: Principles, Practices, and Applications	S. Wesselby	2025	Online course	Practitioner guidance
2	Machine Unlearning: A Comprehensive Survey	Z. Wang et al.	2024	Survey taxonomy	Overview of methods
3	Machine Unlearning: A Survey	H. Xu et al.	2023	ACM Computing Surveys	Establishes unlearning categories
4	MLPro-GT—game theory and dynamic games modeling in Python	S. Yuwono et al.	2024	Framework paper	Open-source GT library

Chapter 3

Research Gaps

Chapter 4

Objectives

Chapter 5

Methodology

Chapter 6

Expected Outcome

Chapter 7

Importance of Proposed Research and SDG Link

Chapter 8

Research Time Plan

Chapter 9

Pilot Study

Chapter 10

Details of Expenses and Source of Funding

Chapter 11

Course Work Details

Chapter 12

References